(ADA) Introduction to the Lee et al. visual cortex recording data

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1 Overview

When modeling brain activations with artificial neural networks (ANNs), previous work found more shared information between early ANN layers and early visual areas and, accordingly, late ANN layers and late visual areas [1–3]. These analyses often report the quality of fit of linear models predicting brain activity from ANN activations [4]. However, as discussed in Feather and Canatar (2023), early and late region/layer data are often not sourced from the same experiment. Thus, these results might be confounded with task differences [5].

In my ADA, I am analyzing dataset provided by Lee et al. (2024) containing longitudinal and inter-region neural responses to visual texture stimuli, available on Zenodo [6]. Lee et al. recorded from 6 macaques with two arrays each, targeting V1, V2, V4, and IT (with the recorded region differing by monkey). All recordings were taken from the same set of texture stimuli.

First, I want to characterize the differences in neural responses for each region and establish differences in encoding of the stimuli. Lee et al. have already performed linear discriminant analysis between noisy/naturalistic textures, and concluded that neural discriminability was highest in V4, lower in V2 and IT, and lowest in V1. For the ADA, I want to characterize encoding differences between texture families. Possible approaches include:

- Multiclass logistic regression (plus some other more detailed interpretation than the inference performance)
- Some non-parametric clustering strategy
- Some distance metric then geometric analysis of the distance matrix

Then, to pursue the question of whether early/late model alignment, possible approaches include:

- Fiting linear models to predict the activity in one region from the activity of another region

- Using V1-like statistics from the provided textures or produce predictors from an established model of V1 firing
- Training an ANN on the noisy/naturalistic or family classification task then comparing the early/late layer alignment

2 Data description

Prior to publishing the data used here, this research group claimed that V2 played a functional role in discriminating noisy vs. naturalistic textures [7]. This follow-up examined how neural and behavioral sensitivity changes over development.

Lee et al. generated stimuli from the Portilla and Simoncelli (PS) parametric texture model [8], producing 15 samples from 35 families at two levels of coherence (noisy vs. naturalistic). They provided four repetitions of each unique image in each session. They implanted two 96-electrode Utah arrays in 6 monkeys.

2.1 Texture stimuli

Example textures generated from the Portilla and Simoncelli (2001) are in Figure 1. The PS parametric texture model is a multilevel generative model that produces sample images by matching visual statistics to an original seed image. The lowest level of the model matches the local statistic of the original image, e.g., the correlation in luminance between adjacent pixels, the average luminance in a small area, or the total variation in luminance. Subsequent layers control higher level correlations, e.g., correlations between neighboring patterns of pixels.

Portilla and Simoncelli describe the higher levels as "V1-like" or analogous to the receptive fields of the primary visual cortex. Matching V1-like correlations produces more coherent *naturalistic* structure. Example noisy and naturalistic images can be seen in Figure 1 (left). Though the degree of naturalistic structure varied in the behavioral task, the passive fixation recordings only contained noisy/zero strength textures or full strength naturalistic textures. Lee et al. generated 15 samples of noisy and naturalistic images from 35 families ($15 \times 35 \times 2$ unique images), where each family corresponded to a different original seed image.

2.2 Passive recording setup

At least two recording sessions were conducted for each monkey: once at a juvenile stage (around 30 weeks) and once at an adult stage (around 56 weeks). Within sessions, images were shown in blocks of 8, with 100 ms of the image on screen followed by 100 ms of blank display. Each of the 1050 unique images (35 families, 15 samples, 2 levels of structure) were presented four times within a session. Lee et al. also included 100 uniquely labeled blank images, also repeated four times. This gives a total of $4 \times (35 \times 15 \times 2 + 100) = 4600$ observations per session.

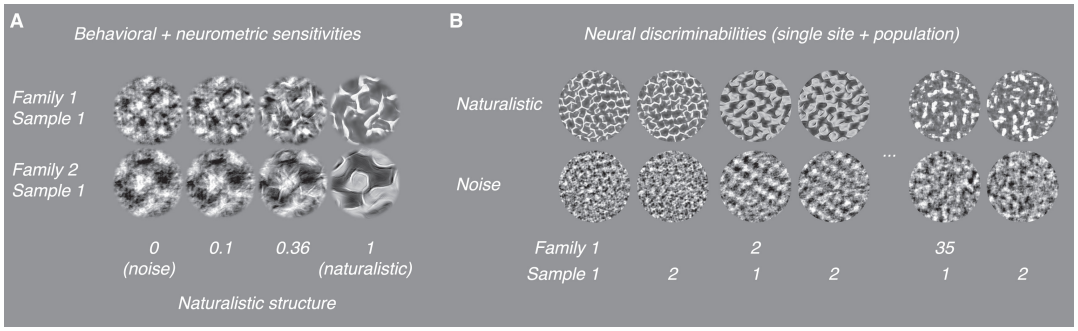


Figure 1: (A) Examples of samples generated with varying levels of naturalistic structure. (B) Families of samples generated from different seed images.

2.3 Recording details

Lee et al. recorded from 6 out of 8 monkeys, with experimental details recopied in Appendix Table 3. The behavioral task is probably not relevant for the ADA, but I included a brief overview in the Appendix Section 6.1. The number of visually responsive sites for each area and session can be seen in Table 1 in Section 3.

The authors implanted a pair of 96 electrode Utah arrays per monkey (i.e., a total of 192 channels per subject). Regions were determined with anatomical landmarks, physiological features (e.g., response latency), and receptive field characteristics. One array implanted on M1 sat on the border between V1 and V2. For M3, the authors could not reliably determine the location and labeled the region as FC (the *foveal confluence* of V1, V2 and V4), based on terminology from Brewer et al. [9]. I removed FC data from analysis.

Lee et al. implanted arrays twice in M6, with the recording at 221 weeks taken from the second implantation. Two monkeys (M4 and M5) only underwent behavioral experiments. One monkey (M8) only underwent passive fixation recording. Three monkeys (M1, M2, and M3) performed the behavioral task prior to passive fixation experiments. Two monkeys (M6 and M7) provided passive fixation recordings prior to undergoing behavioral tests.

The provided Zenodo data files do not contain the original spike timings. Multiunit threshold crossings, or multiunit activations (MUAs), were created from binning spikes into 10 ms windows. Single neuron spikes seem more common in dimensionality analyses [10,11], but other work covers functional dimensionality estimation on fMRI data [12], so this aggregation is probably fine for the purpose of the ADA.

3 Processing multiunit activations

To find visually responsive (reliable) sites, Lee et al. averaged the site response separately for even and odd repetitions of each unique image stimulus (i.e., two groups of two repetitions

each) then kept sites with a Pearson correlation of $\rho \geq 0.4$. Lee et al. defined the onset time for a site as the first time bin with a $|d'| \geq 0.4$ between non-blank and blank stimuli where the two subsequent windows also produced $|d'| \geq 0.4$ of the same sign (see Equation 1 for the definition of d'). They defined the population onset time as the median across visually responsive sites.

$$d' = \frac{\mu_1 - \mu_0}{\sqrt{(\sigma_1^2 + \sigma_0^2)/2}} \tag{1}$$

The procedure for summarizing data can be outlined as:

1. Find a temporary population onset time, with d' calculated between blanks and stimuli. The search window is 0 to 200 ms after stimulus presentation. If no onset time is found, use 100 ms as a placeholder.
2. Determine the responsive sites with signals averaged over a 150 ms window starting at the temporary onset time.
3. Recalculate the population onset time based on the responsive sites.
4. For downstream analyses, sum responses across the 150 ms window after population onset time.

I found fewer reliable recording sites than Lee et al., seen in Table 1 (see Section 5.1 for discussion). Table 2 contain my calculated onset times and the proportion of reliable sites where onset time could be found. As a sanity check, onset times increase from V1 to IT. The proportion of reliable sites with derivable onset time decreases along region hierarchy.

4 Inter-family encoding differences

(No clean work yet).

5 Discussion

5.1 Concerns and assumptions

Stimulus presentation. I assumed that stimuli were presented at index 50 in each MATLAB file based on the averaged responses.

Responsive site calculations. In split-half correlation calculations, I treated the arbitrarily labeled 100 blank images (4 repetitions each) as unique images like the texture images. However, Lee et al. may have omitted the blanks or summarized them, explaining the differences in Table 1.

Aggregating across subjects. The number of recording sites is pretty low, so I aggregated

Table 1: Number of reliable recording sites by region and age, with arbitrarily labeled arrays. Dashes indicate no targeting of the region while zero indicates no active sites detected. No recordings were made in M4 and M5. Outside of the V1/V2 recording in M1, the number of reliable sites are taken out 96 total sites.

	My calculations				Lee et al.			
Age (wks)	Array 1	Array 2	Array 3	Total sites	Array 1	Array 2	Array 3	Total sites
V1								
29	21	NA	NA	21	56	NA	NA	56
37	23	0	NA	23	60	13	NA	73
56	34	NA	NA	34	61	NA	NA	61
V2								
29	19	58	NA	77	34	92	NA	126
37	21	49	NA	70	33	85	NA	118
56	16	45	NA	61	32	92	NA	124
V4								
29	39	3	NA	42	90	21	NA	111
37	43	15	NA	58	91	45	NA	136
56	41	27	NA	68	95	57	NA	152
409	77	NA	NA	77	96	NA	NA	96
IT								
20	13	NA	NA	13	39	NA	NA	39
31	2	4	2	8	40	30	31	101
34	4	2	NA	6	56	48	NA	104
221	11	NA	NA	11	52	NA	NA	52

Table 2: Calculated onset times, when responsive sites are grouped by region and age. The last column reports the proportion of sites where an onset was found.

Age (wks)	Onset time (ms)	Min. onset	Max. onset	Responsive sites	Prop. with onset
V1					
29	40	40	50	21	1.00
37	40	40	50	23	0.957
56	40	40	60	34	1.00
V2					
29	60	40	110	77	0.987
37	60	40	60	70	1.00
56	60	50	110	61	0.984
V4					
29	70	60	150	42	0.810
37	100	60	130	58	0.431
56	70	50	120	68	0.471
409	60	50	60	77	0.987
IT					
20	140	140	140	13	0.0769
31	110	100	130	8	0.375
34	90	90	150	6	0.667
221	90	90	90	11	0.0909

across subjects. For IT, unless the responsive site discrepancy is addressed, I might also need to aggregate across ages.

6 Appendix

6.1 Recording and behavioral task details

Lee et al. measured behavioral sensitivities with a four-choice oddity task. Monkeys were shown 3 noisy distractors (all unique) and 1 naturalistic target, with all four images produced from the same family. To calculate sensitivity curves, Lee et al. varied the strength of the naturalistic structure/V1-like statistic matching. Behavioral sensitivity to naturalistic structure improved with age, though neural sensitivity remained consistent.

Table 3 is copied from Table 1 of Lee et al.

Table 3: Recording ages marked with [a] were used in the comparison in Figure 4D. The column "Behavior" refers to the age range when behavioral tests were administered.

Subject	Behavior (wks)	Areas	Recording ages (weeks; number of sites)
M1 (f)	17–95	V1-V2, V4	29 (n = 56, 34, 90), 37 (n = 60, 33, 91), 56 (n = 61, 32, 95)
M2 (f)	16–95	V2, V4	29 (n = 92, 21), 37[a] (n = 85, 45), 56 (n = 92, 57)
M3 (m)	16–65	FC, IT	20[a] (n = 87, 39), 31 (n = 81, 40), 34 (n = 55, 0 FC only)
M4 (f)	27–53	N/A	N/A
M5 (m)	17–64	N/A	N/A
M6 (m)	184–194	V1, IT	30 (n = 0, 30) 34[a] (n = 0, 56), 36[a] (n = 13, 0)
M6	(see above)	IT	221[a] (n = 52)
M7 (m)	184–194	IT	28 (n = 31), 34[a] (n = 48)
M8 (m)	N/A	V4	409[a] (n = 96)

7 References

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